

TASK 1 — Linux IP Investigation

```
ec2-user@ip-172-31-24-230:~$ ssh -i .Downloads/day1.pem ec2-user@54.167.32.158
Windows PowerShell
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PS C:\Users\u> ssh -i .Downloads/day1.pem ec2-user@54.167.32.158
The authenticity of host '54.167.32.158 (54.167.32.158)' can't be established.
ED25519 key fingerprint is SHA256:ofUAAubc2t51A81sAOm8uzlWTAYb7n825B9LssgyB8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.167.32.158' (ED25519) to the list of known hosts.
```



```
#
##### Amazon Linux 2023
#####
#####|
#####|/
# https://aws.amazon.com/linux/amazon-linux-2023
#####
#####
#####
```

```
[ec2-user@ip-172-31-24-230 ~]$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:ff:e8:33:af:23 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-0c7747b7a456d5522
    altname device-number-0.0
    inet 172.31.24.230/20 metric 512 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 3267sec preferred_lft 3267sec
    inet6 fe80::8ff:e8ff:fe33:af23/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[ec2-user@ip-172-31-24-230 ~]$
```

```
ec2-user@ip-172-31-24-230:~$ ip route show
default via 172.31.16.1 dev ens5 proto dhcp src 172.31.24.230 metric 512
172.31.0.2 via 172.31.16.1 dev ens5 proto dhcp src 172.31.24.230 metric 512
172.31.16.0/20 dev ens5 proto kernel scope link src 172.31.24.230 metric 512
172.31.16.1 dev ens5 proto dhcp scope link src 172.31.24.230 metric 512
ec2-user@ip-172-31-24-230:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
resolv.conf mode: uplink

Link 2 (ens5)
Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    DNS Servers: 172.31.0.2
    DNS Domain: ec2.internal
ec2-user@ip-172-31-24-230:~$ |
```

TASK 2 — IPv4 Address Analysis

```
ec2-user@ip-172-31-24-230:~$ hostname -I
172.31.24.230
[ec2-user@ip-172-31-24-230 ~]$ ping -c 3 8.8.8.8
ping -c 3 1.1.1.1
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=1.57 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=1.53 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=1.55 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 1.533/1.552/1.572/0.015 ms
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=57 time=1.53 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=57 time=1.53 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=57 time=1.53 ms

--- 1.1.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 1.527/1.529/1.531/0.001 ms
[ec2-user@ip-172-31-24-230 ~]$
```

TASK 3 — Dynamic IP Investigation

```
ec2-user@ip-172-31-24-230:~$ curl -s http://169.254.169.254/latest/meta-data/public-ipv4; echo ""
hostname -I

172.31.24.230
[ec2-user@ip-172-31-24-230 ~]$
```


TASK 5 — Cloud Network Troubleshooting

```
ec2-user@ip-172-31-24-230: ~$ sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.

=====
Package                                Architecture      Version            Repository          Size
=====
Installing:
httpd                                  x86_64            2.4.68-1.amzn2023.0.1  amazonlinux         46 k
Installing dependencies:
apr                                    x86_64            1.7.5-1.amzn2023.0.4  amazonlinux         129 k
apr-util                              x86_64            1.6.5-1.amzn2023.0.1  amazonlinux         100 k
apr-util-lmdb                         x86_64            1.6.5-1.amzn2023.0.1  amazonlinux         15 k
generic-logos-httpd                  noarch            18.0.0-12.amzn2023.0.3  amazonlinux         19 k
httpd-core                            x86_64            2.4.68-1.amzn2023.0.1  amazonlinux         1.4 M
httpd-filesystem                     noarch            2.4.68-1.amzn2023.0.1  amazonlinux         12 k
httpd-tools                           x86_64            2.4.68-1.amzn2023.0.1  amazonlinux         80 k
libbrotli                             x86_64            1.0.9-4.amzn2023.0.2  amazonlinux         315 k
mailcap                               noarch            2.1.49-3.amzn2023.0.3  amazonlinux         33 k
Installing weak dependencies:
apr-util-openssl                     x86_64            1.6.5-1.amzn2023.0.1  amazonlinux         17 k
mod_http2                            x86_64            2.0.42-1.amzn2023.0.1  amazonlinux         167 k
mod_lua                               x86_64            2.4.68-1.amzn2023.0.1  amazonlinux         59 k

Transaction Summary
=====
Install 13 Packages

Total download size: 2.4 M
Installed size: 7.0 M
Downloading Packages:
(1/13): apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64.rpm 594 kB/s | 15 kB 00:00
(2/13): apr-util-1.6.5-1.amzn2023.0.1.x86_64.rpm 3.2 MB/s | 100 kB 00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm 3.3 MB/s | 129 kB 00:00
(4/13): apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64.rpm 644 kB/s | 17 kB 00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm 698 kB/s | 19 kB 00:00
(6/13): httpd-2.4.68-1.amzn2023.0.1.x86_64.rpm 1.5 MB/s | 46 kB 00:00
(7/13): httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch.rpm 425 kB/s | 12 kB 00:00
(8/13): httpd-tools-2.4.68-1.amzn2023.0.1.x86_64.rpm 3.4 MB/s | 80 kB 00:00
(9/13): httpd-core-2.4.68-1.amzn2023.0.1.x86_64.rpm 22 MB/s | 1.4 MB 00:00
```

```
ec2-user@ip-172-31-24-230: ~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:ff:e8:33:af:23 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-0c7747b7a456d5522
    altname device-number-0.0
    inet 172.31.24.230/20 metric 512 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 2515sec preferred_lft 2515sec
    inet6 fe80::8ff:e8ff:fe33:af23/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
[ec2-user@ip-172-31-24-230 ~]$ ip route
default via 172.31.16.1 dev ens5 proto dhcp src 172.31.24.230 metric 512
172.31.0.2 via 172.31.16.1 dev ens5 proto dhcp src 172.31.24.230 metric 512
172.31.16.0/20 dev ens5 proto kernel scope link src 172.31.24.230 metric 512
172.31.16.1 dev ens5 proto dhcp scope link src 172.31.24.230 metric 512
[ec2-user@ip-172-31-24-230 ~]$ ping -c 2 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=0.501 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=0.478 ms

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1066ms
rtt min/avg/max/mdev = 0.478/0.489/0.501/0.011 ms
[ec2-user@ip-172-31-24-230 ~]$
```

```
ec2-user@ip-172-31-24-230:~$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-22 19:51:31 UTC; 2min 36s ago
     Docs: man:httpd.service(8)
  Main PID: 4380 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:  0 B/sec"
     Tasks: 177 (limit: 1059)
    Memory: 13.9M
       CPU: 219ms
    CGroup: /system.slice/httpd.service
            └─4380 /usr/sbin/httpd -DFOREGROUND
              └─4473 /usr/sbin/httpd -DFOREGROUND
                └─4483 /usr/sbin/httpd -DFOREGROUND
                  └─4484 /usr/sbin/httpd -DFOREGROUND
                    └─4485 /usr/sbin/httpd -DFOREGROUND

Sep 22 19:51:31 ip-172-31-24-230.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Sep 22 19:51:31 ip-172-31-24-230.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Sep 22 19:51:31 ip-172-31-24-230.ec2.internal httpd[4380]: Server configured, listening on: port 80
[ec2-user@ip-172-31-24-230 ~]$ sudo ss -tuln | grep :80
tcp    LISTEN 0      511      *:80      *:*
[ec2-user@ip-172-31-24-230 ~]$
```

